Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/12/2005
PI Name:	Pickering, Karen D Ph.D.		
Project Title:	Advanced Water Recovery Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	karen.d.pickering@nasa.gov	**Fax:**	FY 281-483-5060
PI Organization Type:	NASA CENTER	**Phone:**	281-483-2688
Organization Name:	NASA - Johnson Space Center		
PI Address 1:	EC3		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	01/01/1998	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	6
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Edeen, Marybeth		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This task provides for technology development for water recovery systems. The task encompasses both biological and physical/chemical system development. Together these technology areas provide technology development for both near and long-term ISS requirements as well as technology development for future extended duration missions. The primary focus is on microgravity-compatibile technologies and ISS-similar wastewater streams; however, early definition of more complex wastewater streams is planned. Included in the biological system development is continued evaluation of bench-scale biological water processors and testing of microgravity-compatible nitrification systems. Physical-chemical system development includes development of high-recovery membrane filtration systems and optimization of postprocessing systems.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Water supplies are a significant consumable in a spacecraft life support system. Efficient technologies for water recovery and reuse greatly reduce the cost of spacecraft resupply. This task develops technology for water recovery systems, including biological systems, membrane filtration systems, and photolytic post-processing systems. Establishing the performance and reliability of these systems is the initial step in reducing the cost of resupply for spacecraft water systems.
Task Progress:	An AWRS technology development project continued this year to study the minimum size anoxic reactor to treat a given flight crew's wastewater for a specific time period. Successful function checkouts were conducted in 2004 and modifications to the nitrification bioreactor and the ultra differential pressure transmitter were made to improve performance. A hydraulic tracer study and aeration study have been completed. The nitrification bioreactor was inoculated in November of 2004. An AWRS study to construct, analyze, and model a gravity independent reactor is also continuing in the Development Facility at the Johnson Space Center. In FY04, test stand construction was completed and certified for operation. An automated data acquistion system was completed to continuously monitor pH, dissolved oxygen, temperature, pressure, and liquid/air differential pressure. Evaluation of the conversion of ammonium to nitrate as a function of residence time and ammonium load was initiated in May of 2004. Within AWRS, work is on-going to collect sufficient data to accurately size a minimum mass water Post Processing System (PPS). A PPS was previously tested as part of the Integrated Advanced Water Recovery System (IAWRS) at the Johnson Space Center. Phase I testing has been completed, with the evaluation of an alternative separate cation/anion ion exchange configuration designed to increase bed life by reducing bicarbonate loading. The performance and bed life of the new configuration was evaluated against a standard mixed-media ion exchange bed configuration. An AWRS effort is underway to evaluate and select catalysts that will convert total organic carbon in reclaimed water to carbon dioxide, eliminating the need for an ion exchange bed, reducing the process power requirements and operational complexity of water recovery systems. In FY04, a novel catalyst from Environmental and Life Support Technologies (ELS) was tested. The catalyst is a carbonized resin, loaded with 5% platinum and ruthenium. Preliminary evalations of the ELS catalyst were encouraging and are currently under evaluation. A low-temperature oxidation catalyst was also tested in FY04. The low-temperature catalyst has proven to be very effective in oxidizing organics from gas streams, such as automotive exhaust. However, testing of this catalyst with an organic laden water stream did not result in the oxidation of organics at a sufficient level to merit further testing. In the area of physical/chemical systems, efforts continued with the Vapor Phase Catalytic Ammonia Removal System. In FY04, a series of tests were conducted to evaluate the system's performance and better understand the micro-gravity compatibility issues currently associated with the hardware. Based upon test observations, modifications were made to the system which greatly improved micro-gravity compatibility.
Bibliography Type:	Description: (Last Updated: 09/21/2006)
Papers from Meeting Proceedings	Muirhead, D.; Potess M.; Rainwater J. "Education Outreach Associated with Technology Transfer in a Colonia of South Texas: Green Valley Farms Science and Space Club for Middle School Aged Children in Green Valley Farms, San Benito, Texas" International Conference on Environmental Systems Jul-2004
Papers from Meeting Proceedings	Finger, B.; Verostko, C.; Carrier, C. "Ersatz Wastewater Formulations for Testing Water Recovery Systems" International Conference on Environmental Systems Jul-2004
Papers from Meeting Proceedings	Vega, L.; Pickering, K.; "Evaluation of Performance of Five Parallel Biological Water Processors" International Conference on Environmental Systems Jul-2004
Papers from Meeting Proceedings	Finger, B. "Optimizing the Ion Exchange Process for Purification of Spacecraft Wastewater" ASCE Earth and Space 2004 Conference Mar-2004
Papers from Meeting Proceedings	Crenwelge, L. "Gas Bubble Dissolution in a Non-venting Gas Trap" ASCE Earth and Space 2004 Conference Mar-2004